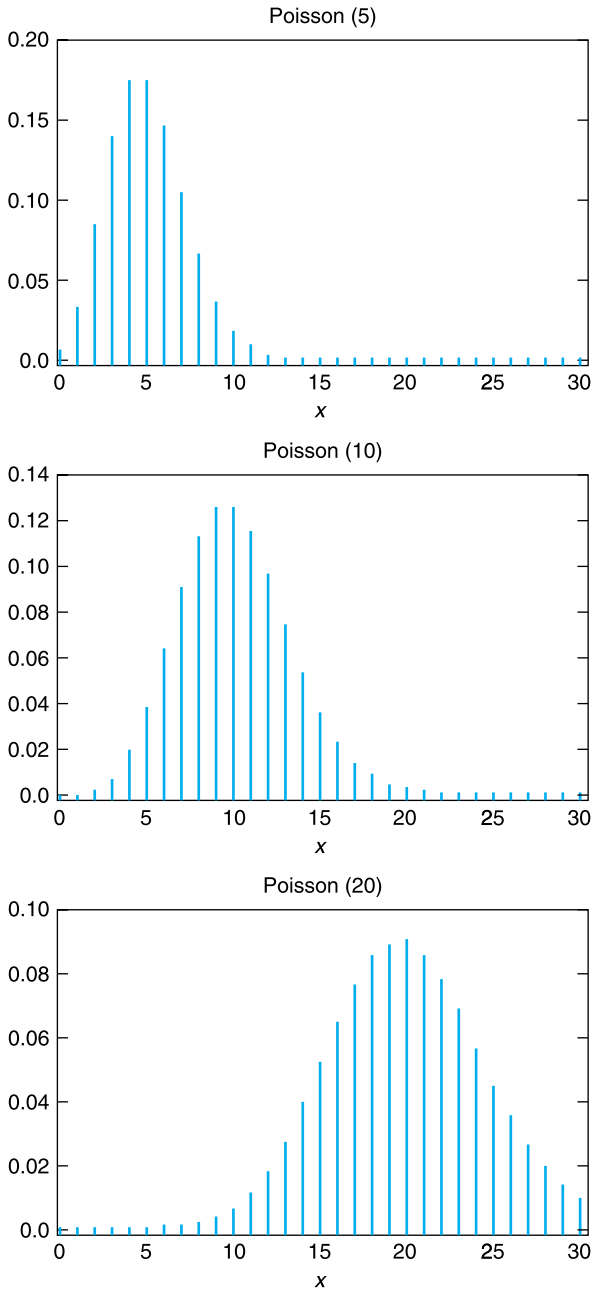


FIGURE 6.4
Poisson probability mass functions.

17. Use the text disk to compute $P\{X \leq 10\}$ when X is a binomial random variable with parameters $n = 100$, $p = .1$. Now compare this with its **(a)** Poisson and **(b)** normal approximation. In using the normal approximation, write the desired probability as $P\{X < 10.5\}$ so as to utilize the continuity correction.
18. The temperature at which a thermostat goes off is normally distributed with variance σ^2. If the thermostat is to be tested five times, find
    a. $P\{S^2/\sigma^2 \leq 1.8\}$
    b. $P\{.85 \leq S^2/\sigma^2 \leq 1.15\}$

    where S^2 is the sample variance of the five data values.
19. In Problem 18, how large a sample would be necessary to ensure that the probability in part (a) is at least .95?
20. Consider two independent samples — the first of size 10 from a normal population having variance 4 and the second of size 5 from a normal population having variance 2. Compute the probability that the sample variance from the second sample exceeds the one from the first. (*Hint*: Relate it to the F-distribution.)
21. Twelve percent of the population is left-handed. Find the probability that there are between 10 and 14 left-handers in a random sample of 100 members of this population. That is, find $P\{10 \leq X \leq 14\}$, where X is the number of left-handers in the sample.
22. Fifty-two percent of the residents of a certain city are in favor of teaching evolution in high school. Find or approximate the probability that at least 50 percent of a random sample of size n is in favor of teaching evolution, when
    a. $n = 10$;
    b. $n = 100$;
    c. $n = 1000$;
    d. $n = 10{,}000$.
23. The following table gives the percentages of individuals of a given city, categorized by gender, that follow certain negative health practices. Suppose a random sample of 300 men is chosen. Approximate the probability that
    a. at least 150 of them rarely eat breakfast;
    b. fewer than 100 of them smoke.

	Sleeps 6 Hours or Less per Night	Smoker	Rarely Eats Breakfast	Is 20 Percent or More Overweight
Men	22.7	28.4	45.4	29.6
Women	21.4	22.8	42.0	25.6

Source: U.S. National Center for Health Statistics, Health Promotion and Disease Prevention.

24. (Use the table from Problem 23.) Suppose a random sample of 300 women is chosen. Approximate the probability that

a. at least 60 of them are overweight by 20 percent or more;
b. fewer than 50 of them sleep 6 hours or less nightly.

25. (Use the table from Problem 23.) Suppose random samples of 300 women and of 300 men are chosen. Approximate the probability that more women than men rarely eat breakfast.
26. The following table uses data concerning the percentages of teenage male and female full-time workers whose annual salaries fall in different salary groupings. Suppose random samples of 1000 men and 1000 women were chosen. Use the table to approximate the probability that
    a. at least half of the women earned less than $20,000;
    b. more than half of the men earned $20,000 or more;
    c. more than half of the women and more than half of the men earned $20,000 or more;
    d. 250 or fewer of the women earned at least $25,000;
    e. at least 200 of the men earned $50,000 or more;
    f. more women than men earned between $20,000 and $24,999.

Earnings Range	Percentage of Women	Percentage of Men
$4999 or less	2.8	1.8
$5000 to $9999	10.4	4.7
$10,000 to $19,999	41.0	23.1
$20,000 to $24,999	16.5	13.4
$25,000 to $49,999	26.3	42.1
$50,000 and over	3.0	14.9

Source: *U.S. Department of Commerce, Bureau of the Census.*

27. Today, roughly 10.5 percent of the labor force belong to a union. If five workers are randomly chosen, what is the probability that none of them belong to a union? Compare your answer to what it would have been in 1983 when 20.1 percent of the workforce belonged to a union.
28. The sample mean and sample standard deviation of all San Francisco student scores on the most recent Scholastic Aptitude Test examination in mathematics were 517 and 120. Approximate the probability that a random sample of 144 students would have an average score exceeding
    a. 507;
    b. 517;
    c. 537;
    d. 550.
29. The average salary of newly graduated students with bachelor's degrees in chemical engineering is $53,600, with a standard deviation of $3200. Approximate the probability that the average salary of a sample of 12 recently graduated chemical engineers exceeds $55,000.

30. A certain component is critical to the operation of an electrical system and must be replaced immediately upon failure. If the mean lifetime of this type of component is 100 hours and its standard deviation is 30 hours, how many of the components must be in stock so that the probability that the system is in continual operation for the next 2000 hours is at least .95?